

BY 532 : Immunotechnology

Course Code : BY 532

Course Name : Immunotechnology

L-T-P-C : 3-0-0-3

Intended for : Elective for B.Tech, M.Tech and Ph.D.

Prerequisite : IC136 - Understanding Biotechnology & its Applications or Consent of faculty member

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Module 1:** Introduction to immunology Cells of Immune system, Organs of immune system, Organization of immune system, Innate and adaptive components, Clonal selection hypothesis, B and T-cell maturation, Antibody, structure, diversity and function, Complement system, Cytokines, chemokines and biological functions.). (8 hours)
- **Module 2:** Immunology of the diseases Tuberculosis, Malaria, HIV, Arthritis, Diabetes, Lupus, Cancer. (6 hours)
- **Module 3:** Immune diagnosis Agglutination assays, ELISA: principles and procedures, In-house ELISA development and procedure (uses, type), Enzyme Electro immune transfer bot (EITB): principles and procedures, Lateral flow devices, Immuno fluorescence diagnostic tools: Principles and uses, ELISPOT, Flow cytometer: Principles and uses. (14 hours)
- **Module 4:** Immunoinformatics Principles, tools and uses of AI and informatics in immunology. (3 hours)
- **Module 5:** Vaccines and therapeutics Introduction to Vaccinology, Peptide Vaccine designing, Chimeric, Multi-epitope, RNA based Vaccine designing, Immune therapeutics, Mabs in therapeutics, Therapeutic peptides. (11 hours)

Textbooks:

1. Punt, Stranford, Jones, Owen, **Kuby Immunology**.

References:

1. Roitt, **Essential Immunology**, 12th Edition.
2. Nicholas Sperelaki, **Cell Physiology Source Book: A Molecular Approach**.
3. Bruce Alberts, Julian Lewis, Alexander Johnson, **Molecular Biology of the Cell**.
4. Editor: Samuel Baron, **Medical Microbiology**, 4th Edition.
5. Strachan, Tom, and Andrew P. Read, **Human Molecular Genetics**, 2nd Edition, John Wiley & Sons Inc., 1999.